

Dynamic analysis of trusses

Special structural elements

ME473 Dynamic finite element analysis of structures

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2025



Where do we stand?

Week	Module	Lecture topic	Mini-projects
1	Linear elastodynamics	Strong and weak forms	
2		Galerkin and FE methods	
3		Systematisation of the procedure	Groups formation
4		3d elements, numerical integration	Project 1 statement
5	Special structural elements	Bars and trusses	

Summary

- Recap weeks 1 to 4
- Eigenvalues and eigenvectors errors bounds
- Trusses in 2d
 - Illustrative example of a 2d truss structure
 - Matlab example of a 2d truss in free vibrations
- Trusses in 3d
 - Matlab example of a 3d truss in free vibrations
 - Ansys example of a transient analysis of a Basketball hoop frame

Recommended readings

- ① Logan, A first course in the finite element method, 6th ed. (chap. 3)
- ② Paz and Leigh, Structural dynamics, 6th ed. (chap. 14)
- ③ Ferreira and Fantuzzi, MATLAB Codes for Finite Element Analysis, 2nd ed. (chap. 4 and 5)

Brief recap of weeks 1 to 4
FEM for elastodynamics of solids

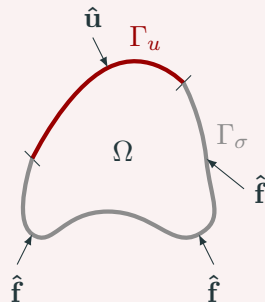
Statement of the linear elastodynamics problem

■ Object:

A solid $\Omega \subset \mathbb{R}^3$ with known material properties: $\mathbf{C}$ and ρ .

■ Main features:

- Acting loads on the body: $\mathbf{f}$.
- Boundary $\Gamma = \Gamma_u \cup \Gamma_\sigma$ (the surface enclosing the solid).
- Boundary conditions: prescribed displacements $\hat{\mathbf{u}}$ on Γ_u and/or loads $\hat{\mathbf{f}}$ on Γ_σ .
- Initial displacement $\mathbf{u}_0$ and velocity $\mathbf{v}_0$ at $t = 0$.



Strong and semi-discrete weak forms of elastodynamics

Strong form

PDE:
$$\nabla^T \mathbf{C} \nabla \mathbf{u} + \mathbf{f} = \rho \ddot{\mathbf{u}}$$

BC on Γ_u :
$$\mathbf{u} = \hat{\mathbf{u}}$$

BC on Γ_σ :
$$\mathbf{N}^T \mathbf{C} \nabla \mathbf{u} = \hat{\mathbf{f}}$$

IC at $t = 0$:
$$\mathbf{u} = \mathbf{u}_0, \dot{\mathbf{u}} = \mathbf{v}_0$$

Semi-discrete weak form

ODE:
$$\delta \mathbf{q}^T [\mathbf{M} \ddot{\mathbf{q}}(t) + \mathbf{K} \mathbf{q}(t) - \mathbf{r}(t)] = 0$$

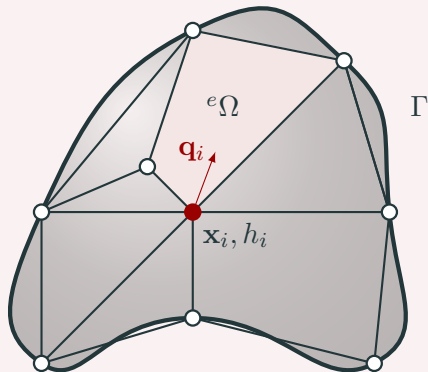
Displacement approximation:
$$\mathbf{u}^h(\mathbf{x}, t) = \mathbf{H}(\mathbf{x}) \mathbf{q}(t) = \sum_{i=1}^p h_i(\mathbf{x}) \mathbf{q}_i(t)$$

IC at $t = 0$:

$$\mathbf{q}(0) = \mathbf{q}_0$$

$$\dot{\mathbf{q}}(0) = \mathbf{p}_0$$

Displacements approximation in finite element method



- Stiffness matrix:

$$\mathbf{K} = \int_{\Omega} \mathbf{B}^T \mathbf{C} \mathbf{B} d\Omega$$

where $\mathbf{B} = \nabla \mathbf{H}$.

- Mass matrix:

$$\mathbf{M} = \int_{\Omega} \rho \mathbf{H}^T \mathbf{H} d\Omega.$$

- Applied forces vector:

$$\mathbf{r}(t) = \int_{\Gamma_\sigma} \mathbf{H}^T \hat{\mathbf{f}} d\Gamma + \int_{\Omega} \mathbf{H}^T \mathbf{f} d\Omega.$$

Types of analysis

Static analysis

Constant loads

Determines the structural deformation $\mathbf{q}$ due to constant loads.

$$\mathbf{K}\mathbf{q} = \mathbf{r}$$

- No inertia effects
- No time dependence
- Equilibrium between internal and external forces

Modal analysis

Natural frequencies

Studies the intrinsic dynamic properties of the structure.

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{0}$$

- No external excitation
- Resonance when excitation frequency $\approx$ natural frequency

Transient analysis

Time-dependent loads

Determines the structural response $\mathbf{q}(t)$ to time-dependent excitations

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{r}(t)$$

- Inertia effects included
- Response evolves in time
- Used for impacts, vibrations and dynamic loading

Eigenvalues and eigenvectors errors bounds

A priori error estimates for eigenvalues and eigenvectors

Using principles from Rayleigh and Courant-Fischer, asymptotic error estimates can be established for eigenvalues and eigenvectors for conformal elements.

$$\lambda_i \leq \lambda_i^h \leq \lambda_i + c h^{2(m-k+1)} \lambda_i^{m+1}$$
$$\|\phi_i^h - \phi_i\|_{L^2} \leq c h^{\min(m+1, 2(m-k+1))} \lambda_i^{(m+1)/2}$$

- λ_i^h are the *approximated* eigenvalues and ϕ_i^h the corresponding eigenvectors,
- λ_i are the *exact* eigenvalues and ϕ_i the corresponding eigenvectors,
- h represents the characteristic mesh size,
- m is the degree of the highest complete polynomial used,
- k is the the highest derivative order appearing in the weak form,
- c is a constant independent of h .

A priori error estimates for frequencies

From the relationship between eigenvalues and frequencies, $\omega_i^h = \sqrt{\lambda_i^h}$, we deduce the bound on the approximate frequencies.

$$\omega_i \leq \omega_i^h \leq \omega_i + c h^{2(m-k+1)} \omega_i^{2m+1}$$

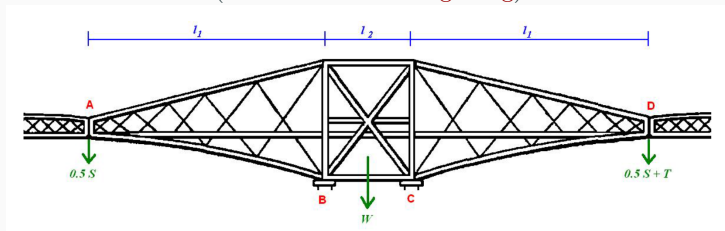
- Approximate frequencies ω_i^h always overestimate the exact frequencies ω_i .
- The quality of approximated frequencies degrades for higher frequencies.
- The convergence rates for eigenvectors and frequencies are both of order h^{2m} for weak forms with $k = 1$ (e.g. elastodynamics).

Structural 1d elements : general ideas

Example: analysis of the forth bridge



(Credit: theforthbridges.org)



(Credit: Chatzi and Egger)

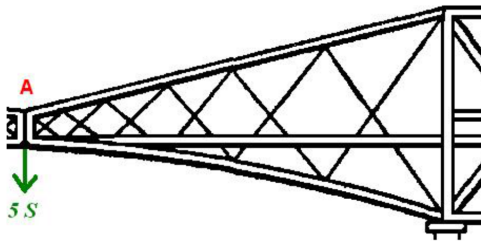
Abstract model

Abstraction

Modelling

Discretization

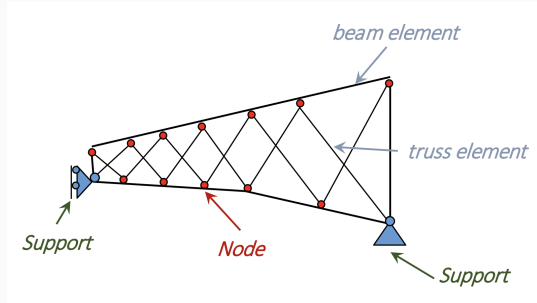
Solution



Conceptual framework: develop a simplified idealisation of the structure, or part of the structure, representing the geometry and loads.

(Credit: Chatzi and Egger)

Mathematical model

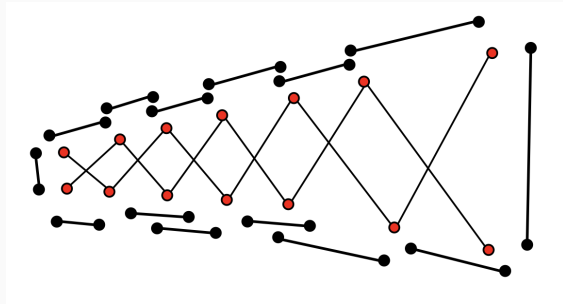


(Credit: Chatzi and Egger)

Connect the structure to a known theoretical model capable of describing its functioning with sufficient accuracy.

Assumption: decompose the system into 1D components, such as trusses and beams.

Discrete model



Direct stiffness method framework:
the continuum is disassembled using a mesh of finite elements that are connected at nodes located on the element boundaries.

(Credit: Chatzi and Egger)

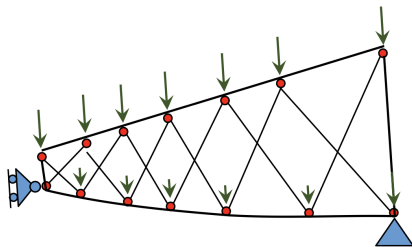
Deformed model

Abstraction

Modelling

Discretization

Solution



(Credit: Chatzi and Egger)

Solve for the structure degrees of freedom:

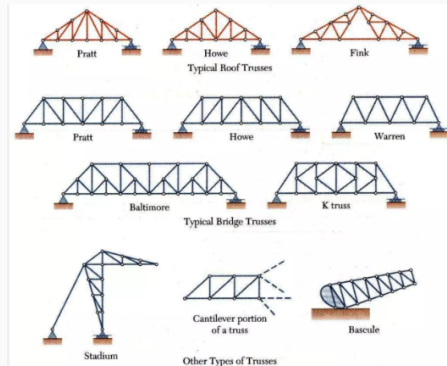
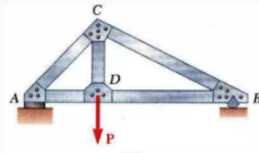
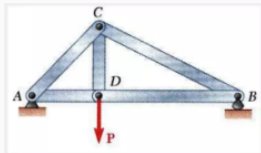
- **assembly** of element quantities and application of boundary conditions,
- **post-processing**: computation of strains, stresses, reaction forces, and moments.

Trusses in 2d

What is a truss structure?

Plane truss

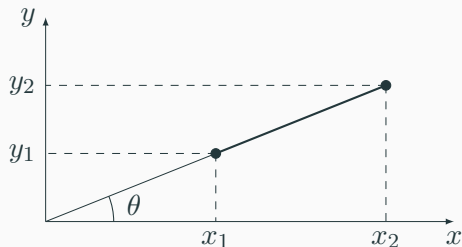
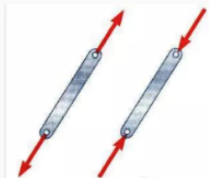
- A structure composed of oriented bar (rod) elements that lie in a common plane and are connected by frictionless pins.
- Loads act only in the common plane and are applied at the nodes or joints.
- Trusses are commonly used in steel buildings, bridges, towers, and other structures.



Kinematic assumptions

Trusses are assumed to exhibit the following characteristics:

- they experience either **compressive** or **tensile forces**,
- their weight is considered negligible in comparison to the loads they support,
- they have varying orientations with respect to a fixed global coordinate system, which serves as a stationary reference framework that remains unchanged regardless of the orientation of individual elements.



Equation of motion for a single non-oriented bar element



- A cross-sectional area
- ℓ length
- E Young's modulus (isotropic bar)
- ρ material density
- u axial displacement
- x' (local) axial coordinate

- Strong form:

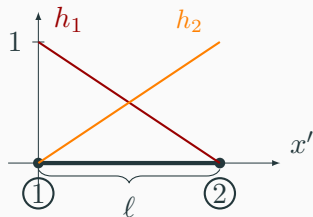
$$\begin{cases} \partial_{x'} (EA \partial_{x'} u(x', t)) = \rho A \ddot{u}(x', t) \\ EA \partial_{x'} u(0, t) = -f_1(t) \\ EA \partial_{x'} u(\ell, t) = f_2(t) \end{cases}$$

$$\begin{cases} u(x', 0) = u_0(x') \\ \dot{u}(x', 0) = v_0(x') \end{cases}$$

- Semi-discrete weak form:

$$\delta \mathbf{q}_{loc}^T [\mathbf{M}_{loc} \ddot{\mathbf{q}}_{loc}(t) + \mathbf{K}_{loc} \mathbf{q}_{loc}(t) - \mathbf{f}_{loc}(t)] = 0$$

Approximated displacements in local coordinates



Linear local shape functions:

$$h_1(x') = 1 - \frac{x'}{\ell}$$

$$h_2(x') = \frac{x'}{\ell}$$

- Displacements approximation local coordinates:

$$u^h(x', t) = h_1(x')q_1(t) + h_2(x')q_2(t) = \begin{bmatrix} h_1(x') & h_2(x') \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix}$$

- Virtual displacements approximation local coordinates:

$$\delta u^h(x') = h_1(x')\delta q_1 + h_2(x')\delta q_2 = \begin{bmatrix} h_1(x') & h_2(x') \end{bmatrix} \begin{bmatrix} \delta q_1 \\ \delta q_2 \end{bmatrix}$$

Elementary quantities in local coordinates

- Element stiffness matrix in local coordinates:

$$\mathbf{K}_{loc} = \int_0^\ell EA \begin{bmatrix} (h'_1)^2 & h'_1 h'_2 \\ h'_2 h'_1 & (h'_2)^2 \end{bmatrix} dx' = \frac{EA}{\ell} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \int_0^\ell \rho A \begin{bmatrix} (h_1)^2 & h_1 h_2 \\ h_2 h_1 & (h_2)^2 \end{bmatrix} dx' = \frac{\rho A \ell}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

- Element applied loads vector in local coordinates:

$$\mathbf{f}_{loc}(t) = \begin{bmatrix} h_1(0) \\ h_2(0) \end{bmatrix} f_1(t) + \begin{bmatrix} h_1(\ell) \\ h_2(\ell) \end{bmatrix} f_2(t) = \begin{bmatrix} f_1(t) \\ f_2(t) \end{bmatrix}$$

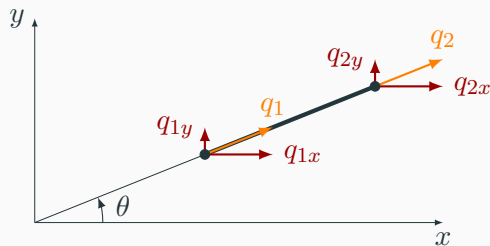
Displacements for a single oriented bar

- Displacement vector in local coordinates:

$$\mathbf{q}_{loc} = [q_1, q_2]^T.$$

- Displacement vector in global coordinates:

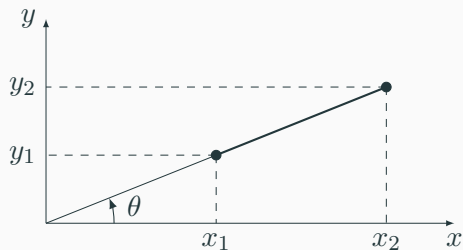
$$\mathbf{q} = [q_{1x}, q_{1y}, q_{2x}, q_{2y}]^T.$$



Relation between local and global displacements:

$$\underbrace{\begin{bmatrix} q_1 \\ q_2 \end{bmatrix}}_{\mathbf{q}_{loc}} = \underbrace{\begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & \cos(\theta) & \sin(\theta) \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} q_{1x} \\ q_{1y} \\ q_{2x} \\ q_{2y} \end{bmatrix}}_{\mathbf{q}}$$

Calculation of direction sines and cosines



The direction sines and cosines can be calculated from the element geometry:

$$\sin(\theta) = \frac{y_2 - y_1}{\ell}$$

$$\cos(\theta) = \frac{x_2 - x_1}{\ell}$$

$$\ell = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Element stiffness and consistent mass matrices in global coordinates

- Element stiffness matrix in global coordinates:

$$\begin{aligned}\mathbf{K} &= \mathbf{T}^T \mathbf{K}_{loc} \mathbf{T} \\ &= \frac{EA}{\ell} \begin{bmatrix} \cos^2(\theta) & \sin(\theta) \cos(\theta) & -\cos^2(\theta) & -\sin(\theta) \cos(\theta) \\ & \sin^2(\theta) & -\sin(\theta) \cos(\theta) & -\sin^2(\theta) \\ & & \cos^2(\theta) & \sin(\theta) \cos(\theta) \\ \text{Symm.} & & & \sin^2(\theta) \end{bmatrix}\end{aligned}$$

- Element consistent mass matrix in global coordinates:

$$\begin{aligned}\mathbf{M} &= \mathbf{T}^T \mathbf{M}_{loc} \mathbf{T} \\ &= \frac{\rho A \ell}{6} \begin{bmatrix} 2 \cos^2(\theta) & 2 \sin(\theta) \cos(\theta) & \cos^2(\theta) & \sin(\theta) \cos(\theta) \\ & 2 \sin^2(\theta) & \sin(\theta) \cos(\theta) & \sin^2(\theta) \\ & & 2 \cos^2(\theta) & 2 \sin(\theta) \cos(\theta) \\ \text{Symm.} & & & 2 \sin^2(\theta) \end{bmatrix}\end{aligned}$$

Lumped mass matrix

- Element consistent mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \frac{\rho A \ell}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

- Element lumped mass matrix in local coordinates:

$$\mathbf{M}_{loc} = \frac{\rho A \ell}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Element lumped mass matrix in global coordinates:

$$\begin{aligned} \mathbf{M} &= \mathbf{T}^T \mathbf{M}_{loc} \mathbf{T} \\ &= \frac{\rho A \ell}{2} \begin{bmatrix} \cos^2(\theta) & \sin(\theta) \cos(\theta) & 0 & 0 \\ \sin(\theta) \cos(\theta) & \sin^2(\theta) & 0 & 0 \\ 0 & 0 & \cos^2(\theta) & \sin(\theta) \cos(\theta) \\ 0 & 0 & \sin(\theta) \cos(\theta) & \sin^2(\theta) \end{bmatrix} \end{aligned}$$

Benefits of using lumped mass matrix

- ✓ **Computational efficiency**

Band matrix for faster computations and lower memory usage.

- ✓ **Improved numerical stability**

Help avoid non-physical coupling between DOFs, which can cause instabilities, in explicit time integration (Newmark or central difference methods).

- ✓ **Physical realism for trusses**

Truss mass is mostly at joints so lumped mass better reflects reality.

- ✓ **Good approximation in practice**

Accurate enough for natural frequency and mode shape estimation.

Band mass matrix = faster, simpler, and often accurate enough!

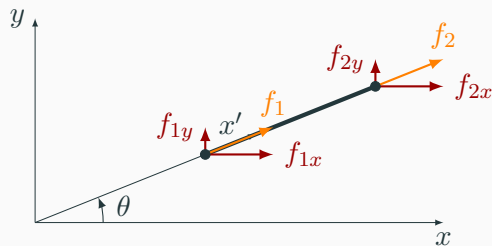
Applied loads for a single oriented bar

- Nodal force vector in local coordinates:

$$\mathbf{f}_{loc} = [f_1, f_2]^T.$$

- Nodal force vector in global coordinates:

$$\mathbf{f} = [f_{1x}, f_{1y}, f_{2x}, f_{2y}]^T.$$



Forces undergo transformation in the same manner as displacements:

$$\underbrace{\begin{bmatrix} f_1 \\ f_2 \end{bmatrix}}_{\mathbf{f}_{loc}} = \underbrace{\begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & \cos(\theta) & \sin(\theta) \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} f_{1x} \\ f_{1y} \\ f_{2x} \\ f_{2y} \end{bmatrix}}_{\mathbf{f}}$$

Elementary loads vector in global coordinates

- Element applied loads vector in global coordinates:

$$\mathbf{f} = \mathbf{T}^T \mathbf{f}_{loc} = \begin{bmatrix} \cos(\theta) f_1 \\ \sin(\theta) f_1 \\ \cos(\theta) f_2 \\ \sin(\theta) f_2 \end{bmatrix}$$

- Loads are only applied at pins and are given in the global coordinates system, the assembled loads vector can be computed directly.
- Distributed/self-weight loads are transformed to equivalent nodal loads.

Assembly of stiffness and mass matrices and loads vector

Given a 2d truss structure made of m oriented bars, n nodes.

1. Element quantities:

- For each bar element e , compute the element quantities global coordinates:

$${}^e\mathbf{K} = {}^e\mathbf{T}^T {}^e\mathbf{K}_{loc} {}^e\mathbf{T}$$

$${}^e\mathbf{M} = {}^e\mathbf{T}^T {}^e\mathbf{M}_{loc} {}^e\mathbf{T}$$

$${}^e\mathbf{f} = {}^e\mathbf{T}^T {}^e\mathbf{f}_{loc}$$

2. Global assembly:

- Initialize global stiffness matrix $\mathbf{K}$ and global mass matrix $\mathbf{M}$ of size $2n \times 2n$.
- Initialize global loads vector $\mathbf{f}$ of size $2n \times 1$.
- Assemble each ${}^e\mathbf{K}$, ${}^e\mathbf{M}$ and ${}^e\mathbf{f}$ for $e = 1, \dots, m$, into $\mathbf{K}$, $\mathbf{M}$ and $\mathbf{f}$ respectively using element connectivity.

Reduced stiffness and mass matrices and loads vector

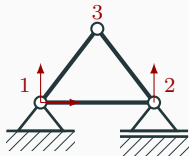
3. Boundary conditions:

- Identify constrained (*fixed or supported*) degrees of freedom,
- Partition global matrices and vectors to separate free and constrained DOFs:

$$\mathbf{K} = \begin{bmatrix} \mathbf{K}_{ff} & \mathbf{K}_{fc} \\ \mathbf{K}_{cf} & \mathbf{K}_{cc} \end{bmatrix}, \quad \mathbf{M} = \begin{bmatrix} \mathbf{M}_{ff} & \mathbf{M}_{fc} \\ \mathbf{M}_{cf} & \mathbf{M}_{cc} \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} \mathbf{f}_f \\ \mathbf{f}_c \end{bmatrix}.$$

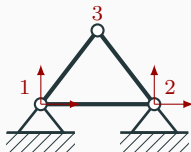
- Apply constraints by removing or modifying rows and columns corresponding to constrained DOFs.

Pin-roller



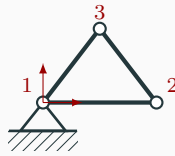
Pin: $q_{1x} = q_{1y} = 0$;
roller: $q_{2y} = 0$.

Pin-pin



Pin: $q_{1x} = q_{1y} = 0$;
Pin: $q_{2x} = q_{2y} = 0$.

Pin-free



Pin: $q_{1x} = q_{1y} = 0$.

Modal analysis

- Solve the free vibrations (homogeneous) problem without external loads:

$$\mathbf{M}_{ff}\ddot{\mathbf{q}}_f(t) + \mathbf{K}_{ff}\mathbf{q}_f(t) = \mathbf{0}.$$

- Assume harmonic motion $\mathbf{q}_f(t) = \boldsymbol{\phi} e^{i\omega t}$ and derive the eigenvalue problem:

$$(\mathbf{K}_{ff} - \omega^2 \mathbf{M}_{ff}) \boldsymbol{\phi} = \mathbf{0}.$$

- Solve for eigenvalues $\lambda = \omega^2$ (squared natural frequencies):

$$\det(\mathbf{K}_{ff} - \lambda \mathbf{M}_{ff}) = 0.$$

- Solve for corresponding eigenvectors $\boldsymbol{\phi}$ (mode shapes). Normalize eigenvectors with respect to $\mathbf{M}_{ff}$ or $\mathbf{K}_{ff}$.

Transient analysis

- The dynamic equilibrium equation for the free DOFs becomes:

$$\mathbf{M}_{ff}\ddot{\mathbf{q}}_f(t) + \mathbf{K}_{ff}\mathbf{q}_f(t) = \mathbf{f}_f(t).$$

- This coupled system of equations can be uncoupled by transforming to modal coordinates using the normal mode matrix Φ , where $\mathbf{q}_f(t) = \Phi\mathbf{z}(t)$.
- Substituting into the equation and pre-multiplying by Φ^T , we obtain a decoupled system:

$$\ddot{\mathbf{z}}(t) + \Lambda\mathbf{z}(t) = \Phi^T\mathbf{f}_f(t),$$

where Λ is a diagonal matrix of squared natural frequencies.

- The decoupled equations in modal space can be solved independently using time integration methods, greatly simplifying the dynamic analysis.

Post-processing: stress computation

- In the local coordinate system, the approximated axial stress in element e is

$${}^e\sigma_{loc}^h = E^e \varepsilon_{loc}^h$$

- Stain-displacement relationship:

$${}^e\varepsilon_{loc}^h = \frac{d}{dx'} u^h = \begin{bmatrix} \frac{dh_1}{dx'} & \frac{dh_2}{dx'} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \frac{1}{e\ell} \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$$

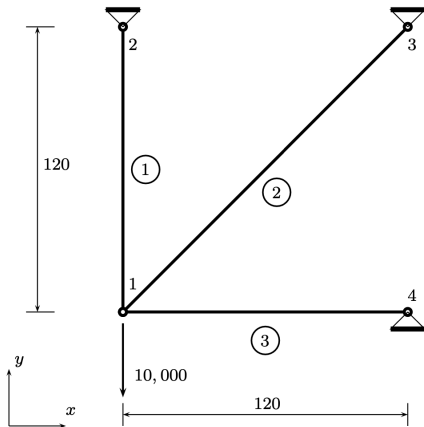
- Using the coordinate transformation ${}^e\mathbf{q}_{loc} = {}^e\mathbf{T}\mathbf{q}$ it leads to the approximated stress computed in global coordinates:

$$\begin{aligned} {}^e\sigma^h &= \frac{{}^eE}{e\ell} \begin{bmatrix} -\cos({}^e\theta) & -\sin({}^e\theta) & \cos({}^e\theta) & \sin({}^e\theta) \end{bmatrix} \begin{bmatrix} q_{ix} \\ q_{iy} \\ q_{jx} \\ q_{jy} \end{bmatrix} \\ &= \frac{{}^eE}{e\ell} [\cos({}^e\theta)(q_{jx} - q_{ix}) + \sin({}^e\theta)(q_{jy} - q_{iy})] \end{aligned}$$

Note that element e is connected to nodes numbered as i and j .

Illustrative example of a 2d truss structure

Problem description



Magnesium alloy material properties:

- cross-sectional area $A = 78.5 \text{ mm}^2$
- Young's modulus $E = 40 \cdot 10^3 \text{ MPa}$
- Density $\rho = 1.810 \text{ ton/m}^3$

Parameters:

Elements	Nodes	$e\theta$	$e\ell$
1	1, 2	90°	120 mm
2	1, 3	45°	$120\sqrt{2} \text{ mm}$
3	1, 4	0°	120 mm

Objective: Compute the equation of motion in semi-discrete weak form.

Credit: Ferreira and Fantuzzi, MATLAB Codes for Finite Element Analysis

Element matrices in global coordinates

■ Element stiffness matrices

$${}^1\mathbf{K} = {}^1\mathbf{T}^T \mathbf{K}_{\text{loc}} {}^1\mathbf{T} = \frac{EA}{1\ell} \begin{bmatrix} \cos(90^\circ) & 0 \\ \sin(90^\circ) & 0 \\ 0 & \cos(90^\circ) \\ 0 & \sin(90^\circ) \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} \cos(90^\circ) & \sin(90^\circ) & 0 & 0 \\ 0 & 0 & \cos(90^\circ) & \sin(90^\circ) \end{bmatrix} = \frac{78500}{3} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}$$

By analogy:

$${}^2\mathbf{K} = \frac{78500}{6\sqrt{2}} \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{bmatrix} \quad {}^3\mathbf{K} = \frac{78500}{3} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

■ Element consistent mass matrices

$${}^1\mathbf{M} = \frac{28417}{10} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \end{bmatrix} \quad {}^2\mathbf{M} = \frac{28417}{20} \sqrt{2} \begin{bmatrix} 2 & 2 & 1 & 1 \\ 2 & 2 & 1 & 1 \\ 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 \end{bmatrix} \quad {}^3\mathbf{M} = \frac{28417}{10} \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Assembly of the global stiffness and consistent mass matrices

Local index	bar 1	bar 2	bar 3
1	1	1	1
2	2	3	4

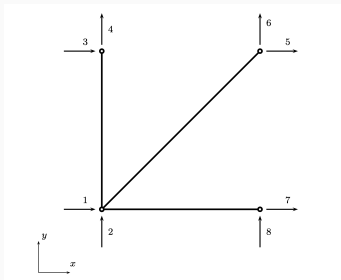
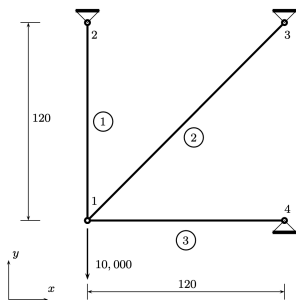
Global stiffness matrix : $K = \frac{78500}{6}$

$$\begin{bmatrix} 2 + 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 & -1/\sqrt{2} & -1/\sqrt{2} & -2 & 0 \\ 1/\sqrt{2} & 2 + 1/\sqrt{2} & 0 & -2 & -1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 2 & 0 & 0 & 0 & 0 \\ -1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ -1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 & 1/\sqrt{2} & 1/\sqrt{2} & 0 & 0 \\ -2 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Global mass matrix : $M = \frac{28417}{20}$

$$\begin{bmatrix} 4 + 2\sqrt{2} & 2\sqrt{2} & 0 & 0 & \sqrt{2} & \sqrt{2} & 2 & 0 \\ 2\sqrt{2} & 4 + 2\sqrt{2} & 0 & 2 & \sqrt{2} & \sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 4 & 0 & 0 & 0 & 0 \\ -\sqrt{2} & -\sqrt{2} & 0 & 0 & 2\sqrt{2} & 2\sqrt{2} & 0 & 0 \\ \sqrt{2} & \sqrt{2} & 0 & 0 & 2\sqrt{2} & 2\sqrt{2} & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Applied loads and boundary conditions



$$\mathbf{f} = \begin{bmatrix} 0 \\ -10000 \\ f_{2x} \\ f_{2y} \\ f_{3x} \\ f_{3y} \\ f_{4x} \\ f_{4y} \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} q_{1x} \\ q_{1y} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

■ The free DOFs are q_{1x} and q_{1y} .

The equations of motions for free nodes are given by :

$$\frac{28417}{20} \begin{bmatrix} 4 + 2\sqrt{2} & 2\sqrt{2} \\ 2\sqrt{2} & 2\sqrt{2} \end{bmatrix} \begin{bmatrix} \ddot{q}_{1x} \\ \ddot{q}_{1y} \end{bmatrix} + \frac{78500}{6} \begin{bmatrix} 2 + 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 2 + 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} q_{1x} \\ q_{1y} \end{bmatrix} = \begin{bmatrix} 0 \\ -10000 \end{bmatrix}$$

Credit: Ferreira and Fantuzzi, MATLAB Codes for Finite Element Analysis

An example of a 2d truss in free vibration

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Trusses in 3d

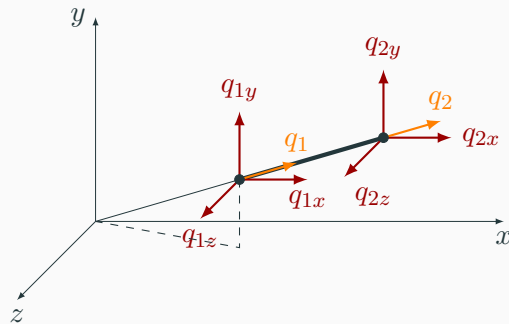
Displacements for oriented bar in 3d

- Displacements in local coord. does not change w.r.t the one in 2d:

$$\mathbf{q}_{loc} = [q_1, q_2]^T.$$

- Displacements in global coordinates projected from nodes 1 and 2 have now 6 components:

$$\mathbf{q} = [q_{1x}, q_{1y}, q_{1z}, q_{2x}, q_{2y}, q_{2z}]^T.$$



Relation between local and global displacements

The relationship between local and global displacements is given by the *direction cosines matrix*:

$$\underbrace{\begin{bmatrix} q_1 \\ q_2 \end{bmatrix}}_{\mathbf{q}_{loc}} = \underbrace{\begin{bmatrix} l_x & l_y & l_z & 0 & 0 & 0 \\ 0 & 0 & 0 & l_x & l_y & l_z \end{bmatrix}}_{\mathbf{T}} \underbrace{\begin{bmatrix} q_{1x} \\ q_{1y} \\ q_{1z} \\ q_{2x} \\ q_{2y} \\ q_{2z} \end{bmatrix}}_{\mathbf{q}}$$

where

$$l_x = \frac{x_2 - x_1}{e\ell}, \quad l_y = \frac{y_2 - y_1}{e\ell}, \quad l_z = \frac{z_2 - z_1}{e\ell}$$

$$e\ell = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Element stiffness matrix in global coordinates

$$\begin{aligned} {}^e\mathbf{K} &= {}^e\mathbf{T}^T \mathbf{K}_{loc} {}^e\mathbf{T} \\ &= \frac{e(EA)}{e\ell} \begin{bmatrix} l_x & l_y & l_z & 0 & 0 & 0 \\ 0 & 0 & 0 & l_x & l_y & l_z \end{bmatrix}^T \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} l_x & l_y & l_z & 0 & 0 & 0 \\ 0 & 0 & 0 & l_x & l_y & l_z \end{bmatrix} \\ &= \frac{e(EA)}{e\ell} \begin{bmatrix} l_x^2 & l_x l_y & l_x l_z & -l_x^2 & -l_x l_y & -l_x l_z \\ & l_y^2 & l_y l_z & -l_x l_y & -l_y^2 & -l_y l_z \\ & & l_z^2 & -l_x l_z & -l_y l_z & -l_z^2 \\ & & & l_x^2 & l_x l_y & l_x l_z \\ & \text{Symm.} & & & l_y^2 & l_y l_z \\ & & & & & l_z^2 \end{bmatrix} \end{aligned}$$

Element mass matrices in global coordinates:

- Consistent mass matrix:

$${}^e\mathbf{M} = {}^e\mathbf{T}^T \mathbf{M}_{loc} {}^e\mathbf{T} = \frac{{}^e(\rho A \ell)}{6} \begin{bmatrix} 2l_x^2 & 2l_x l_y & 2l_x l_z & l_x^2 & l_x l_y & l_x l_z \\ 2l_x l_y & 2l_y^2 & 2l_y l_z & l_x l_y & l_y^2 & l_y l_z \\ 2l_x l_z & 2l_y l_z & 2l_z^2 & l_x l_z & l_y l_z & l_z^2 \\ l_x^2 & l_x l_y & l_x l_z & 2l_x^2 & 2l_x l_y & 2l_x l_z \\ l_x l_y & l_y^2 & l_y l_z & 2l_x l_y & 2l_y^2 & 2l_y l_z \\ l_x l_z & l_y l_z & l_z^2 & 2l_x l_z & 2l_y l_z & 2l_z^2 \end{bmatrix}$$

- Lumped mass matrix:

$${}^e\mathbf{M} = {}^e\mathbf{T}^T \mathbf{M}_{loc} {}^e\mathbf{T} = \frac{{}^e(\rho A \ell)}{2} \begin{bmatrix} l_x^2 & l_x l_y & l_x l_z & 0 & 0 & 0 \\ l_x l_y & l_y^2 & l_y l_z & 0 & 0 & 0 \\ l_x l_z & l_y l_z & l_z^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & l_x^2 & l_x l_y & l_x l_z \\ 0 & 0 & 0 & l_x l_y & l_y^2 & l_y l_z \\ 0 & 0 & 0 & l_x l_z & l_y l_z & l_z^2 \end{bmatrix}$$

An example of a 3d truss in free vibration

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